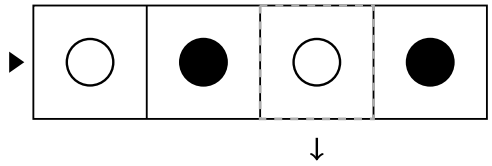
Keep the arrow at the left. A key gives each picture a different row, all the same length. The sender hides the picture and its row. A changer secretly turns over zero or one counter. Only the final row crosses the folder. The receiver knows the key.

Sender and changer

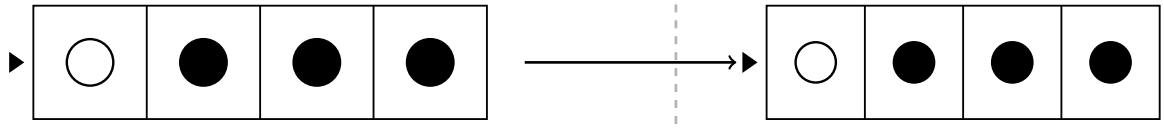
Receiver

Start row



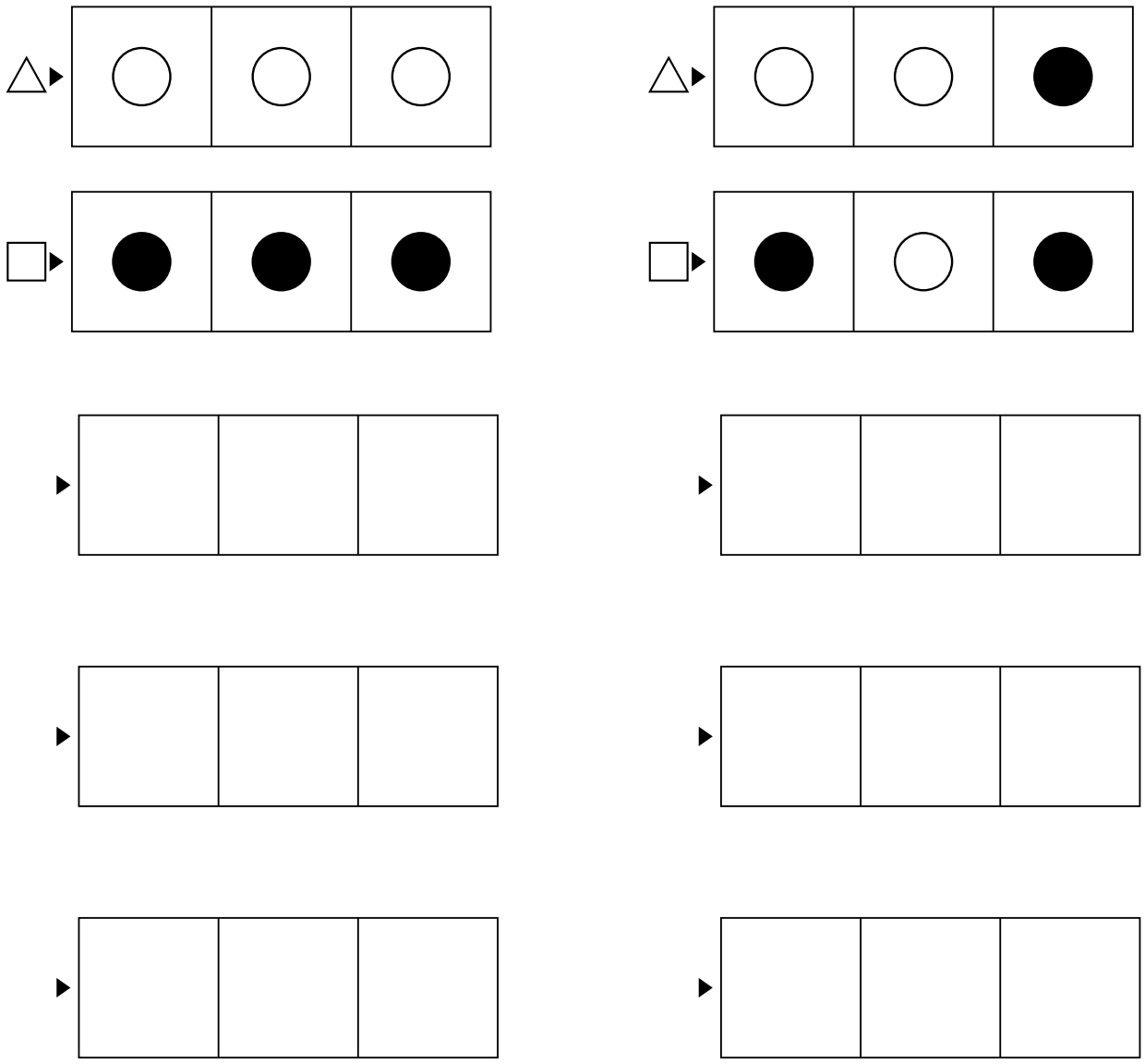
Turn over only the third counter.

Final row

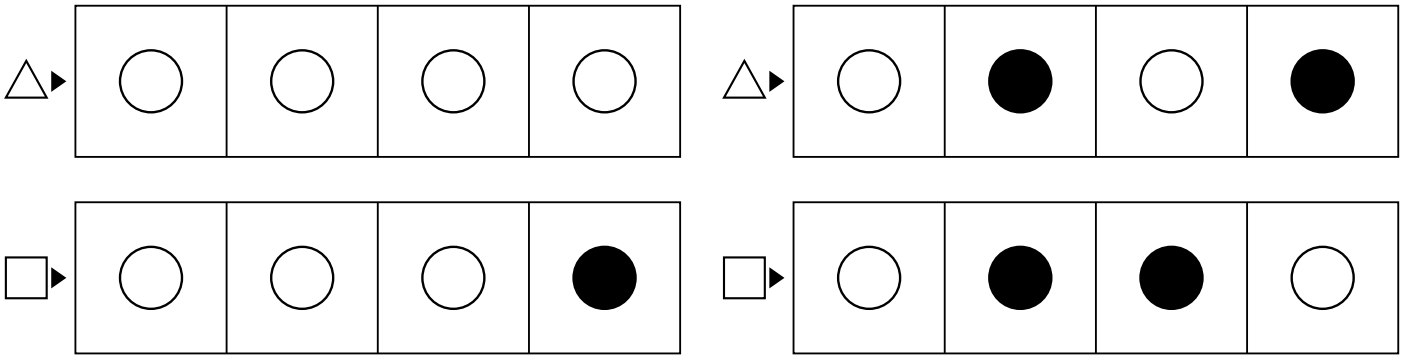


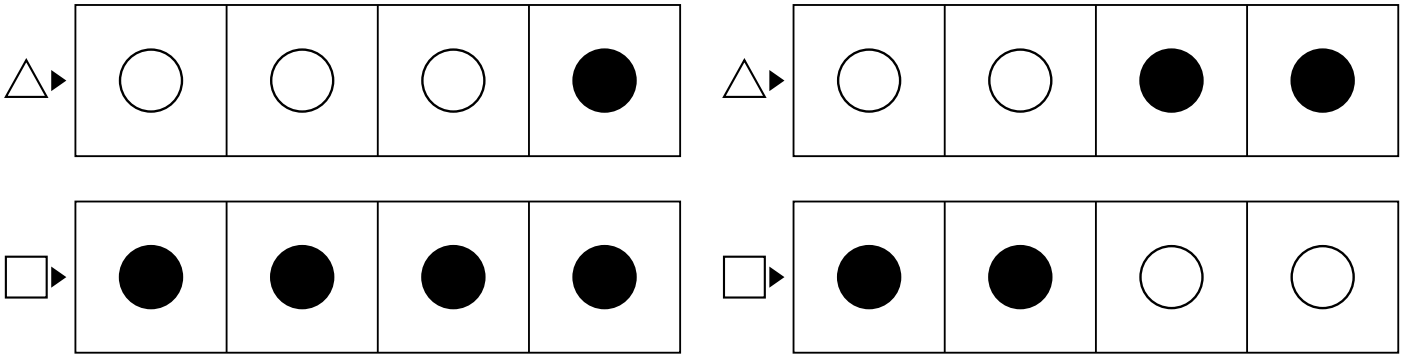
In the game, the start row and the changed position stay hidden behind the folder.

Problem 1: Use each key to send both pictures several times. Which keys always let the receiver tell which picture was sent?

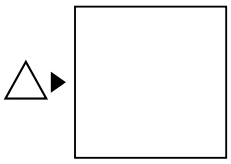


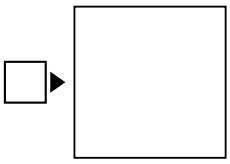
Problem 2: For each key, find a row that the receiver could get from either picture. If there is no such row, explain why.

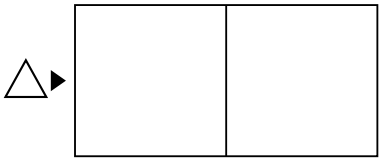


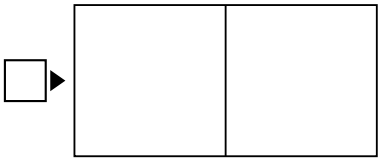


Problem 3: Make a key for the two pictures with 1, 2, or 3 counters in each row. What is the shortest length that always works? Explain why no shorter key works.

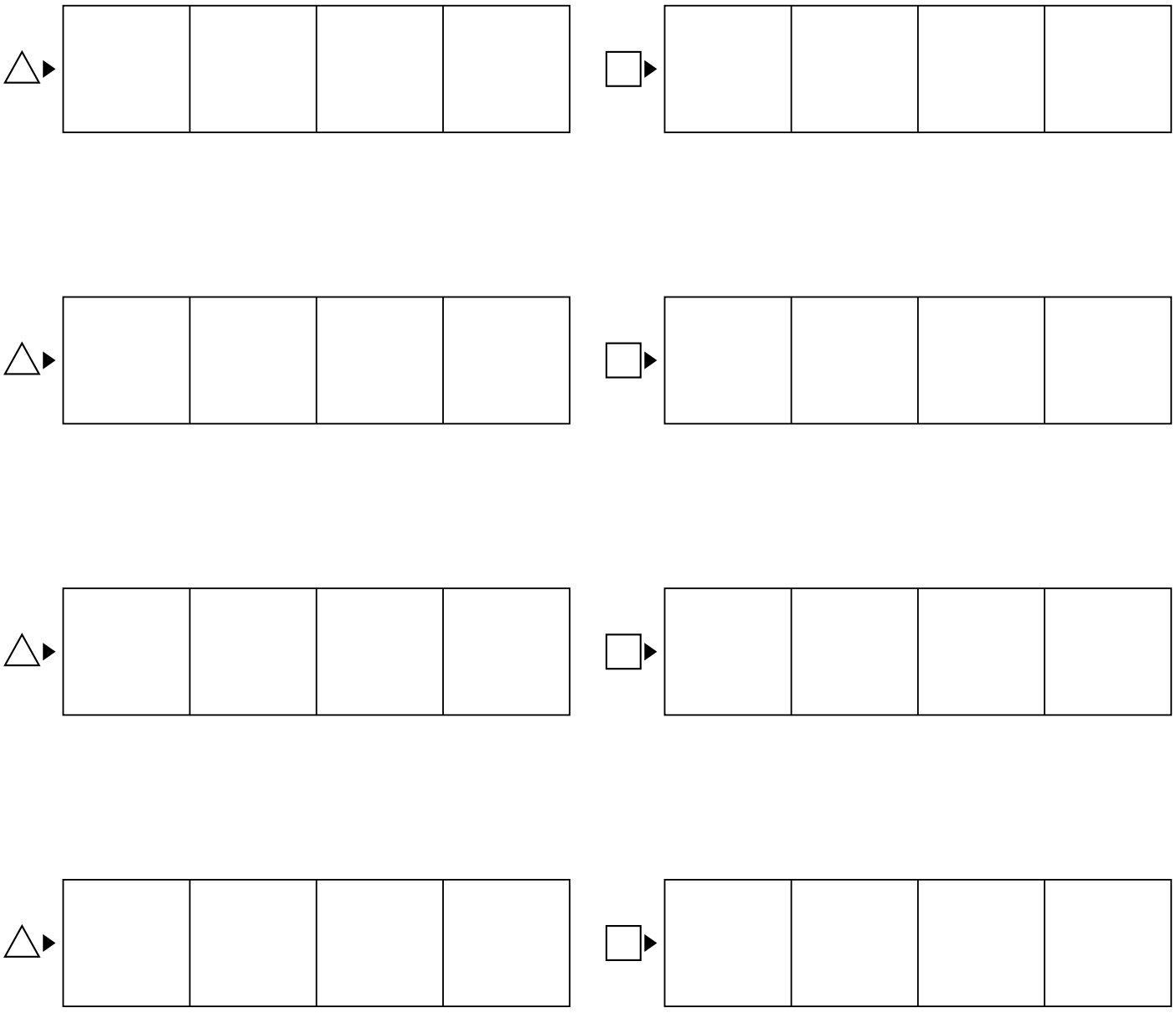




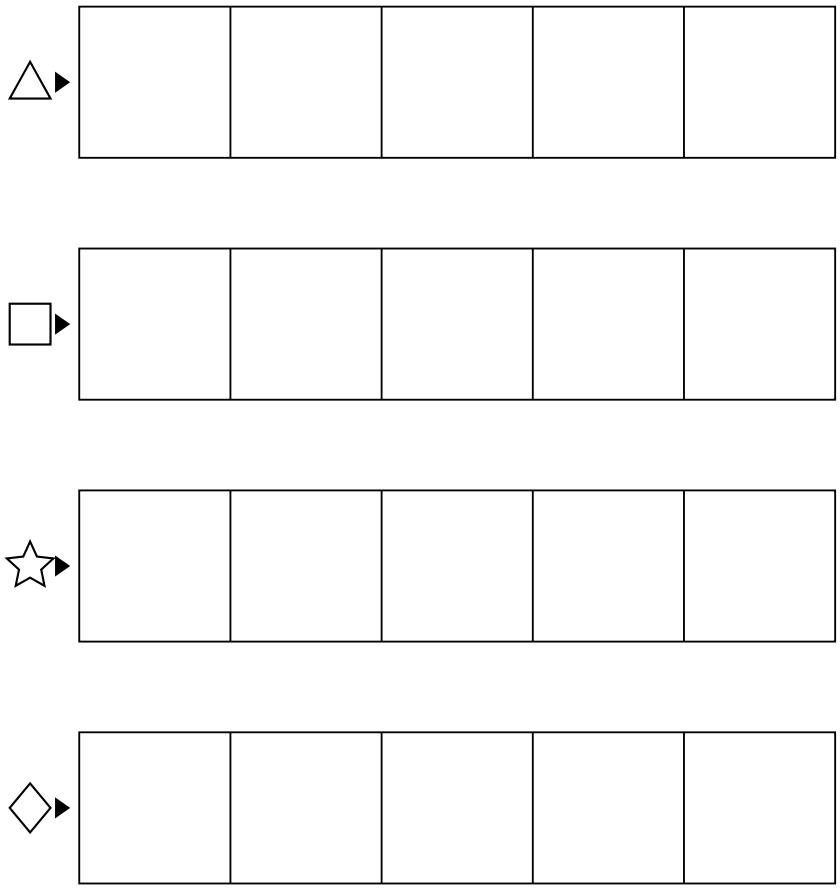




Problem 4: Make four-counter keys for the two pictures with exactly two filled circles in each row. Find every key that always works. Swapping the pictures counts as a different key. Use more paper if needed.



Problem 5: Make a key for all four pictures, with five counters in each row, that always lets the receiver know the picture. Explain how you know your key works.



Problem 6: This time the changer may turn over zero, one, or two counters. Make the shortest key you can for the two pictures, using at most five counters in each row. Explain why your length is the shortest.